

CS5281: INTERNET APPLICATION DEVELOPMENT

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Internet Application Development

Subject Code

CS - Computer Science

Course Number

5281

Academic Unit

Computer Science (CS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

This course provides an overview of the platforms, technologies and architectures used in modern Web-based application development. The objective of this course is to ensure that all students have a good balance of practical hands-on development experience as well as systems-level concepts to ensure that they can make intelligent analysis and critique of platforms and technologies for future development work.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if DEC-A1 DEC-A2 DEC-A3 app.)			
1	Compare, analyze and explain different Web architectures.		x		
2	Explain what are Web standards and their roles and importance in Web development.		x		
3	Create practical website design with consideration of user requirement.			x	x
4	Propose enhancements over static webpages with client-side script/tools.			x	x
5	Propose designs of server-side programs which interact with client-side elements.			x	x
6	Explain current trends in Web technologies and development.		x		

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lecture/Tutorial	Students will explain key concepts, such as client - server architecture, web standards and markup languages.	1, 2, 3, 4, 5, 6	3hrs/wk
2	Group Project	Students will design and implement fully functional internet based system for real-life application like hotel booking or online bidding.	1, 2, 3, 4, 5, 6	on average 3~5hrs/wk

Additional Information for LTAs

Lecture/Tutorial-CILO No. 1-6 (indirectly)

Group Project-CILO No. 1-2, 6 (indirectly)

Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Midterm Quiz	1, 2, 3, 4, 6	15	-	No
2	Group Project	1, 2, 3, 4, 5, 6	35	-	Yes

Continuous Assessment (%)

50

Examination (%)

50

Examination Duration (Hours)

2

Minimum Examination Passing Requirement (%)

30

Additional Information for ATs

For a student to pass the course, at least 30% of the maximum mark for the examination must be obtained.

Assessment Rubrics (AR)

Assessment Task

Midterm Quiz (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

1.1 ABILITY to EXPLAIN and compare different Internet technologies with focus placed on client side development.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Group Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

2.1 ABILITY to DESIGN and implement web-based Internet application.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

3.1 ABILITY to EXPLAIN and compare different Internet technologies for both client side and server side.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Midterm Quiz (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

1.1 ABILITY to EXPLAIN and compare different Internet technologies with focus placed on client side development.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Group Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

2.1 ABILITY to DESIGN and implement web-based Internet application.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

3.1 ABILITY to EXPLAIN and compare different Internet technologies for both client side and server side.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal levels

Part III Other Information**Keyword Syllabus**

History of Internet and the Web, overview of Web-based architectures and platforms, client-server model, browser and web server structures, Web standards and protocols (HTTP, XML, HTML, CSS), client-side programming (JavaScript), server-side programming (e.g. PHP / Java platform, AJAX, Web services), Framework for PC/Mobile platform (e.g. JQuery), Web 2.0 (concept and technologies).

Reading List**Compulsory Readings**

Title	
1	None - Freely available online material will be used

Additional Readings

Title	
1	Nil